# Rose Polyhouse Misting Controller – Proposal and Layout

Oct 31, 2026 · @Santosh Kumar Jha

**Agri Sensors and Controls** (formerly Sasya Systems) · agrisenseandcontrol.in

Agri Sensors and Controls will automate misting in the 2,800 m² rose polyhouse with two FG1 units: a controller at the pump that measures outside temperature and humidity and switches the existing 7 HP starter, and a sensor node in the middle of the house. More sensor nodes can be added later by plug-and-play, and the same kit repeats for the second polyhouse and its own pump. It runs on its own without internet; everything is set from the ASAC FG1 mobile app.

## Scope (phase 1)

Phase 1 controls the misting pump only: the FG1 relay switches the existing L&T starter directly.

- **FG1 Controller** (1 no.), in the pump panel: relay to the existing starter, its own temperature/humidity sensor for outside (shaded) conditions, and the only unit connected to the internet.
- **FG1 Sensor Node** (1 no.), at the centre of the polyhouse: same FG1 hardware with temperature/humidity sensor, no relay.
- Controller and node talk to each other directly by wireless; no router is needed between them.
- More sensor nodes can be added any time: power on, add in the app, done.
- Automatic misting by temperature and humidity, in timed ON/OFF pulses (ON and OFF times adjustable), only inside a daily working window (start and end times adjustable).
- Choose which sensor(s) control misting: one node, an average of selected nodes, or an average of all.
- Manual ON/OFF from the app, plus an Auto/Manual selector switch on the panel so the pump can always be run the old way.
- ASAC FG1 mobile app shows every node's readings: on site through the controller's own WiFi hotspot, and from anywhere through farm WiFi with internet.
- Readings and misting history, with alerts on the phone.

# Sensor nodes and control input

Every FG1 in a polyhouse reports temperature and humidity to that house's controller, and the grower chooses in the app which readings drive the misting.

| Control input option | When to use it |
|---|---|
| One selected node (default: centre node) | Simplest; the centre of the house represents the crop best |
| Average of selected nodes | Ignore a node in an unrepresentative spot (near a door or the pad wall) |
| Average of all nodes | Large house with several nodes spread evenly |

- If a selected node stops reporting, the controller uses the remaining selected nodes; if none are left, misting stops and an alert is sent.
- The controller's own sensor sits in the shaded pump-panel area and measures outside temperature and humidity. The app shows it as "Outside" next to the inside readings, so the grower sees how much cooler and more humid the house is kept. It is not part of the control input by default, but can be added.

**Adding a node (plug and play)**

1. Mount the new node and switch on its 5 V adapter.
2. In the app, the controller shows "New node found"; tap **Add**, give it a name (e.g. "North end").
3. Choose whether it is part of the control input. Its readings and history appear at once.

A node only joins the controller it was added to, so nodes of neighbouring polyhouses never mix.

**Second polyhouse (3,300 m², with its own pump)**

- Has its own pump and gets the same kit: its own FG1 Controller switches that pump's starter and sits in the same pump panel as Controller 1, about 45 m from polyhouse 2; its sensor node(s) go inside polyhouse 2.
- Both houses appear in one farm in the app, each with its own settings, readings and history.
- Wireless range: the polyhouse 2 node must reach its controller about 45 m away, through poly film and crop. We will test this at the site visit during phase 1. If the signal is weak, the options are an external antenna on the controller, a node placed midway as a relay, or the long-range radio (LoRa) version of the node.

# How it decides

Inside the working window (11:00–15:00 to start, both times adjustable), misting starts when the house is hot or dry, runs in ON/OFF pulses (2 min ON / 3 min OFF to start, both adjustable), and stops as soon as it is cool or humid enough; the stop rule always has priority.

| Setting | Starting value | Adjustable in app |
|---|---|---|
| Start misting: temperature at or above | 32 °C | Yes |
| Start misting: humidity at or below (only when temperature is above the stop temperature) | 45 % RH | Yes |
| Stop misting: temperature at or below | 29 °C | Yes |
| Stop misting: humidity at or above | 60 % RH | Yes |
| Pump ON time per pulse | 2 min | Yes |
| Pump OFF time per pulse | 3 min | Yes |
| Working window (misting only between) | 11:00 to 15:00 | Yes |

These are starting values for the field trial and will be fine-tuned with the grower. With 2 min ON / 3 min OFF the pump starts about 12 times per hour while misting; the controller enforces a minimum OFF time so it can never restart too quickly.

**Examples (inside the 11:00–15:00 window)**

| Reading in the house | Result | Why |
|---|---|---|
| 33 °C, 50 % RH | Misting | Temperature at or above 32 °C |
| 31 °C, 42 % RH | Misting | Humidity at or below 45 % and temperature above 29 °C |
| 33 °C, 62 % RH | No misting | Humidity at or above 60 %: stop has priority |
| 30 °C, 65 % RH | No misting | Humidity at or above 60 % |
| 28 °C, 40 % RH | No misting | Temperature at or below 29 °C |
| 31 °C, 52 % RH | Continues as it was | Between the start and stop limits, so no change |
| 34 °C, 50 % RH at 16:00 | No misting | Outside the working window |

In short: whenever humidity is 60 % or more, there is no misting at any temperature.

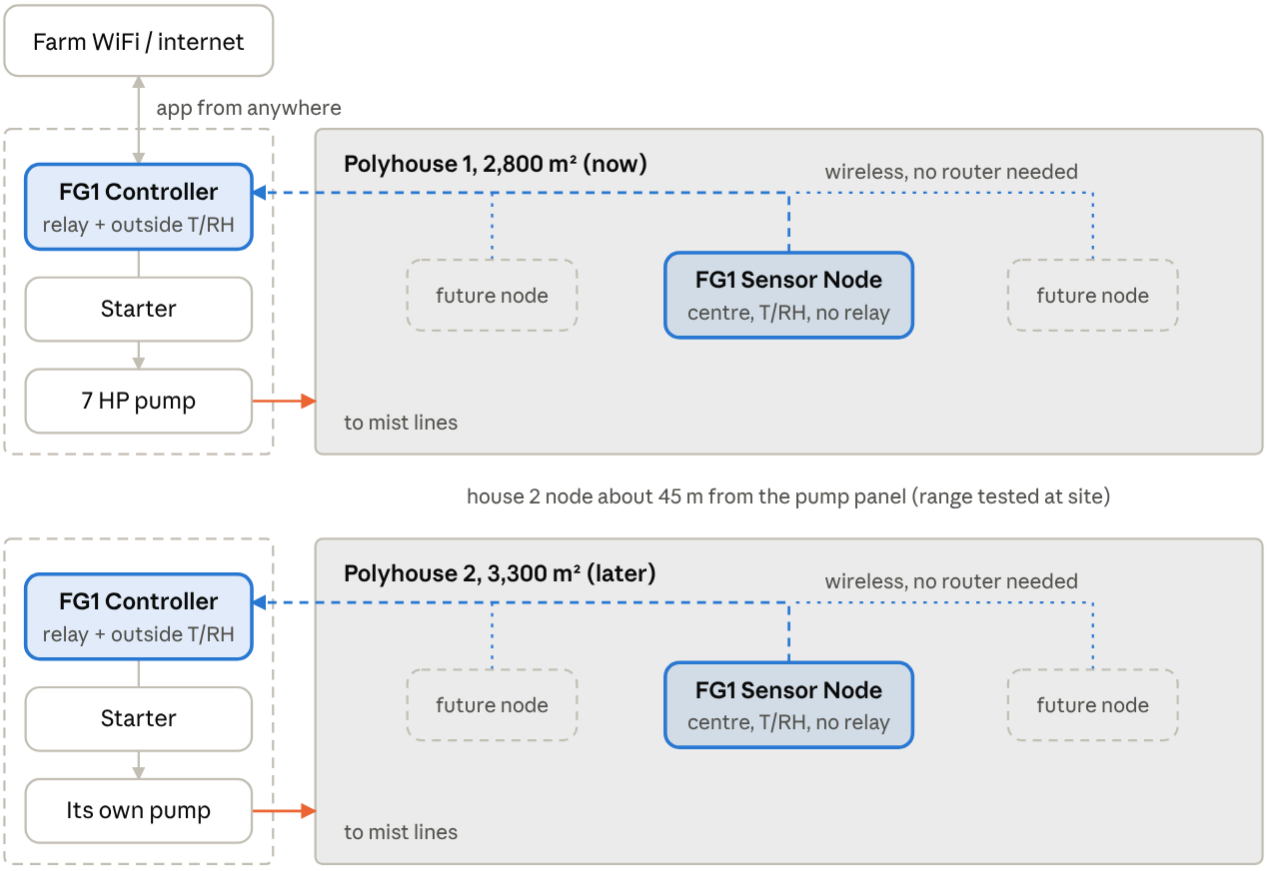
**Other behaviour**

- Readings are averaged over 1 minute so a passing gust does not switch the pump.
- If all sensors in the control input stop reporting for 2 minutes, misting stops and an alert is sent.
- After a power cut the controller comes back in Auto mode with all settings kept.
- Manual ON from the app switches back to Auto after a set time (default 30 min).

## Layout in the polyhouse

Each polyhouse has one FG1 Controller at its pump and one or more FG1 Sensor Nodes inside; nodes talk wirelessly to their own controller, and only the controller connects to the internet.

**One controller per house in the pump panel; nodes talk only to it**




Plan view, not to scale. Final node positions are fixed at the site visit.
Each node is added to one controller only, so the two houses never mix readings.
Both controllers sit in the same shaded pump panel and measure outside conditions; both reach farm WiFi.

polyhouse layout · 2 sensors, 1 controller

Sensor nodes hang at plant canopy height inside their shields, away from doors, side walls and direct spray: a wet sensor reads 100 % humidity and would stop misting wrongly. Future nodes can go at about one-third and two-thirds of the house length. If the wireless range from the house to the pump shed is weak, a node placed between them extends it; this is checked at the site visit.

## Specifications

### FG1 Controller (1 no. per polyhouse, in the pump panel)

| Item | Specification |
|---|---|
| Processor | ESP32, WiFi 2.4 GHz + Bluetooth; same temperature/humidity sensor as the nodes, in a shield, measuring outside conditions in the shaded panel area |
| Output | 1 relay, SPDT, contacts rated 10 A at 250 V AC; switches the starter coil circuit |
| Clock | Real-time clock with coin-cell backup; keeps the working window correct without internet |
| Power | 220 V AC single phase, through a 5 V 1 A adapter |
| Enclosure | IP65 box, 105 × 105 × 80 mm, cable glands |
| Indicators | Power/WiFi LED, pump-ON LED |
| Spare input | Flow-sensor input, for dry-run protection later |
| Memory | All settings and recent history kept through power cuts |
| Connectivity | Own WiFi hotspot for on-site app use; farm WiFi + internet (MQTT) for remote use; firmware updates from the app |

### FG1 Sensor Node (1 no. now, more any time)

| Item | Specification |
|---|---|
| Hardware | Same FG1 board as the controller, without the relay |
| Sensor | Digital temperature and humidity sensor, Sensirion SHT40 class (about ±0.2 °C, ±1.8 % RH) |

| Item | Specification |
|---|---|
| Housing | Sensor inside a ventilated multi-plate radiation shield, so sun and spray do not affect the reading |
| Reading interval | Every 10 s |
| Link to controller | Direct wireless (ESP-NOW, 2.4 GHz) to its own controller; works with no router and no internet |
| Adding | Plug and play: power on, then Add in the app |
| Power | 5 V adapter from a 220 V supply at the node (provided by the customer at every node) |
| Mounting | On a post or crop wire at plant canopy height, height adjustable as the crop grows |

### Mobile app (ASAC FG1, Android)

| Screen | What it shows or does |
|---|---|
| Farm | Both polyhouses side by side, each with its current temperature, humidity and pump state |
| Dashboard | Live temperature and humidity from every node inside, plus the outside reading from the controller, pump ON/OFF, time left in the current ON or OFF period, reason for misting |
| Sensors | Add a new node, rename it, see its signal; choose the control input: one node, average of selected nodes, or average of all |
| Control | Auto / Manual ON / Manual OFF |
| Settings | All values in the "How it decides" table; protected by a PIN |
| History | Temperature and humidity graph per node, with misting periods marked; list of pump starts and stops |
| Alerts | Node not reporting, very high temperature, controller offline |

# From the customer, installation and later options

### Please arrange

- [ ] A 220 V single-phase supply in the pump panel for the controllers
- [ ] A 220 V supply at every node position: the centre of each polyhouse now, and any future node position later
- [ ] Farm WiFi name and password, if remote use is wanted (WiFi must reach the pump panel)
- [ ] Phone numbers of the people who should get the app and alerts
- [ ] Someone on site during installation and the first trial days

### Installation steps

1. Site visit: confirm polyhouse size, sensor positions, panel wiring and wireless range from the house to the panel.
2. Fit the controller and the Auto/Manual switch in the pump panel; wire to the starter.
3. Mount the sensor node, add it in the app, choose the control input; check all readings against a reference thermometer.
4. Test in Manual mode, then in Auto with the starting values.
5. Field trial for 1–2 weeks: tune temperature, humidity, ON/OFF times and window with the grower.
6. Handover and app training for the owner and staff.

### Later options (quoted separately)

- Second polyhouse (3,300 m²): the same Controller + Node kit for its own pump, shown in the same app (see "Sensor nodes and control input").
- Dry-run protection with a flow sensor on the mist line.
- Phase-failure relay in the starter circuit.
- WhatsApp/SMS alerts and a weekly climate report.
- Control of fans, shade net or vents from a multi-relay controller.
- Mist by VPD (a combined heat-and-humidity stress value) instead of separate limits.

---

**Agri Sensors and Controls** (formerly Sasya Systems) · https://agrisenseandcontrol.in/